E-commerce Performance Analysis

Project Report

Project Title: E-commerce Performance Dashboard
Tools Used: Microsoft Excel
Analysis Period: January to June 2026
Dataset: E-commerce performance Dataset
Currency: Egyptian Pound (EGP)

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1. Executive Summary

This project focuses on analyzing the performance of a fictional e-commerce business over a six-month period, from January to June 2026.

The objective was to transform raw e-commerce data into meaningful business insights using Microsoft Excel. The analysis covers sales performance, product categories, customer acquisition channels, purchase conversion, and customer retention.

An interactive-style dashboard was developed to present the key performance indicators (KPIs), visualizations, and findings in a clear and accessible format.

The analysis recorded total revenue of **133,895 EGP** from **165 orders**, with an average order value of **811.48 EGP**. The overall session-to-purchase conversion rate was **17.17%**.

2. Project Objectives

The main objectives of this project were to:

- Analyze monthly revenue and identify sales trends.
- Evaluate revenue contribution across product categories.
- Compare the performance of customer acquisition channels.
- Examine the customer journey through the e-commerce conversion funnel.
- Analyze customer retention using cohort analysis.
- Build an Excel dashboard that presents the findings in a clear and organized way.

3. Dataset Overview

The project uses a synthetic dataset created for educational and analytical purposes. It represents a fictional e-commerce business and contains information about users, website sessions, and customer orders.

The workbook includes the following main datasets:

Users

- User ID

- Signup Date
- Acquisition Channel
- Country
- Device

Sessions

- Session ID
- User ID
- Session Date
- App Open
- View Product
- Add to Cart
- Begin Checkout
- Purchase

Orders

- Order ID
- Session ID
- User ID
- Order Date
- Product Name
- Product Category
- Quantity
- Unit Price
- Discount Percentage
- Revenue
- Acquisition Channel

The dataset covers the period from January to June 2026.

4. Key Performance Indicators (KPIs)

The dashboard highlights four main performance indicators:

KPI	Result
Total Revenue	133,895 EGP
Total Orders	165
Average Order Value (AOV)	811.48 EGP
Purchase Conversion Rate	17.17%

KPI Definitions:

- **Total Revenue:** The sum of revenue generated from all recorded orders.
- **Total Orders:** The total number of recorded orders.
- **Average Order Value (AOV):** Total revenue divided by the total number of orders.
- **Purchase Conversion Rate:** The number of sessions that resulted in a purchase divided by the total number of recorded sessions.

5. Sales Performance Analysis

5.1 Monthly Revenue Analysis

Monthly revenue was analyzed to identify changes in sales performance throughout the six-month period.

Month	Revenue (EGP)
January 2026	17,185.50
February 2026	16,149.00
March 2026	31,416.50
April 2026	31,999.50
May 2026	20,978.50
June 2026	16,166.00

Key Finding:

Revenue peaked in April 2026 at **31,999.50 EGP**, while June recorded the lowest monthly revenue at **16,166 EGP**.

The data shows a noticeable increase in revenue during March and April, followed by a decline in May and June.

5.2 Revenue by Product Category

This analysis compares the revenue generated by each product category.

Product Category	Revenue (EGP)
Fashion	45,563.00
Electronics	30,708.50
Home	19,476.00
Accessories	17,007.00
Sports	10,642.50
Beauty	10,498.00

Key Findings:

- Fashion was the highest-revenue category, generating 45,563 EGP.
- Electronics ranked second with 30,708.50 EGP.
- Beauty recorded the lowest revenue at 10,498 EGP.
- Fashion generated approximately 4.34 times the revenue of Beauty.

5.3 Revenue by Acquisition Channel

Revenue was also analyzed across the different customer acquisition channels.

Acquisition Channel	Revenue (EGP)
Organic Search	36,844.00
Paid Ads	27,467.50
Referral	27,387.50
Social Media	26,460.50
Email	15,735.50

Key Findings:

- Organic Search generated the highest revenue at 36,844 EGP.
- Paid Ads and Referral generated similar revenue.
- Email recorded the lowest revenue among the five channels.

These results show the revenue contribution of each channel. However, revenue alone does not establish which channel is the most cost-effective, as acquisition and marketing costs were not included in the dataset.

6. Conversion Funnel Analysis

The conversion funnel was used to examine how sessions progressed through the purchasing journey.

Funnel Stage	Sessions	Conversion from Previous Stage	Drop-off Rate
App Open	961	—	—
View Product	961	100%	0%
Add to Cart	365	38%	62%
Begin Checkout	212	58%	42%
Purchase	165	78%	22%

Key Findings:

- 961 sessions opened the application, and all 961 recorded sessions also viewed a product.
- 365 sessions proceeded to add a product to the cart.
- 212 sessions reached the checkout stage.
- 165 sessions resulted in a purchase.
- The largest drop-off occurred between **View Product and Add to Cart**, where 62% of sessions did not proceed to the next stage.

Business Implication:

The transition from viewing a product to adding it to the cart represents the largest opportunity for further investigation. The business could examine product information, pricing, product availability, and user experience to better understand this drop-off.

These are potential areas for investigation, not confirmed causes established by the dataset.

7. Customer Cohort and Retention Analysis

Cohort analysis was conducted to understand how customer activity changed over time after signup.

Users were grouped according to their signup month. Their activity was then tracked across subsequent months.

The retention heatmap displays the percentage of users in each cohort who remained active during each observed month.

Key Findings:

- The April 2026 cohort recorded the highest first-month retention rate at **56%**.
- First-month retention for the other cohorts ranged from 44% to 52%.
- The January 2026 cohort recorded 45% retention in Month 1, declining to 21% by Month 5.

- Retention percentages generally declined as more time passed after signup.

Business Implication:

The retention analysis provides a way to compare customer activity across signup cohorts. The April cohort's first-month retention result may be worth investigating further to identify whether similar engagement patterns can be achieved in other cohorts.

Blank cells in the heatmap represent future months that had not yet been observed in the dataset, rather than zero retention.

8. Dashboard Overview

The final Excel dashboard brings together the main results of the analysis in one visual report.

It includes:

- **KPI Cards:** Total Revenue, Total Orders, Average Order Value, and Purchase Conversion Rate.
- **Monthly Revenue Trend:** A line chart showing revenue changes from January to June 2026.
- **Revenue by Product Category:** A horizontal bar chart comparing category performance.
- **Revenue by Acquisition Channel:** A horizontal bar chart comparing revenue across acquisition channels.
- **E-commerce Conversion Funnel:** A visualization of session progression through the purchasing stages.
- **Cohort Retention Heatmap:** A visual comparison of customer retention across signup cohorts.
- **Key Insights:** A summary of the most important findings from the analysis.

The dashboard was designed to make the results easier to understand and to support a quick overview of the business's performance.

9. Tools and Techniques Used

Microsoft Excel

The project was completed using Microsoft Excel, with the following features and techniques:

- Excel Tables for organizing structured datasets.
- Formulas for calculating KPIs and analytical metrics.
- PivotTables for summarizing and analyzing data.
- Charts for visualizing sales and customer behavior.
- Conditional Formatting for the cohort retention heatmap.
- Data organization and formatting to create a structured dashboard.

10. Recommendations

Based on the analysis, the following areas could be investigated further:

1. **Product Page Engagement:** Investigate why a large percentage of sessions that viewed products did not add items to their carts.
2. **Product Category Performance:** Explore the factors contributing to Fashion's higher revenue and identify opportunities in lower-performing categories.
3. **Acquisition Channel Performance:** Investigate the effectiveness of each acquisition channel using additional metrics such as marketing costs and customer acquisition cost.
4. **Customer Retention:** Study the engagement patterns of different signup cohorts and explore ways to improve repeat activity.
5. **Sales Trends:** Investigate the changes in monthly revenue, particularly the decline observed after April.

These recommendations are based on the patterns observed in the available dataset. Additional data would be required to confirm the underlying causes and evaluate the impact of any proposed actions.

11. Project Limitations

There are several limitations to consider when interpreting the results:

- The dataset is synthetic and does not represent actual business transactions or customer behavior.
- The analysis covers only six months.
- Marketing expenditure and customer acquisition costs were not available.
- The conversion funnel is based on recorded sessions, not unique customers.
- Cohort retention is based on the observed activity records available in the dataset.
- The analysis identifies patterns and potential areas for investigation but does not establish causal relationships.

12. Conclusion

This project demonstrates how Microsoft Excel can be used to organize data, calculate business KPIs, analyze sales performance, examine customer conversion behavior, and evaluate customer retention.

By combining structured data analysis with visual reporting, the project transforms raw e-commerce data into a dashboard that communicates key findings in a clear and practical way.

It also provided practical experience in Excel-based data analysis, business performance measurement, and data visualization.